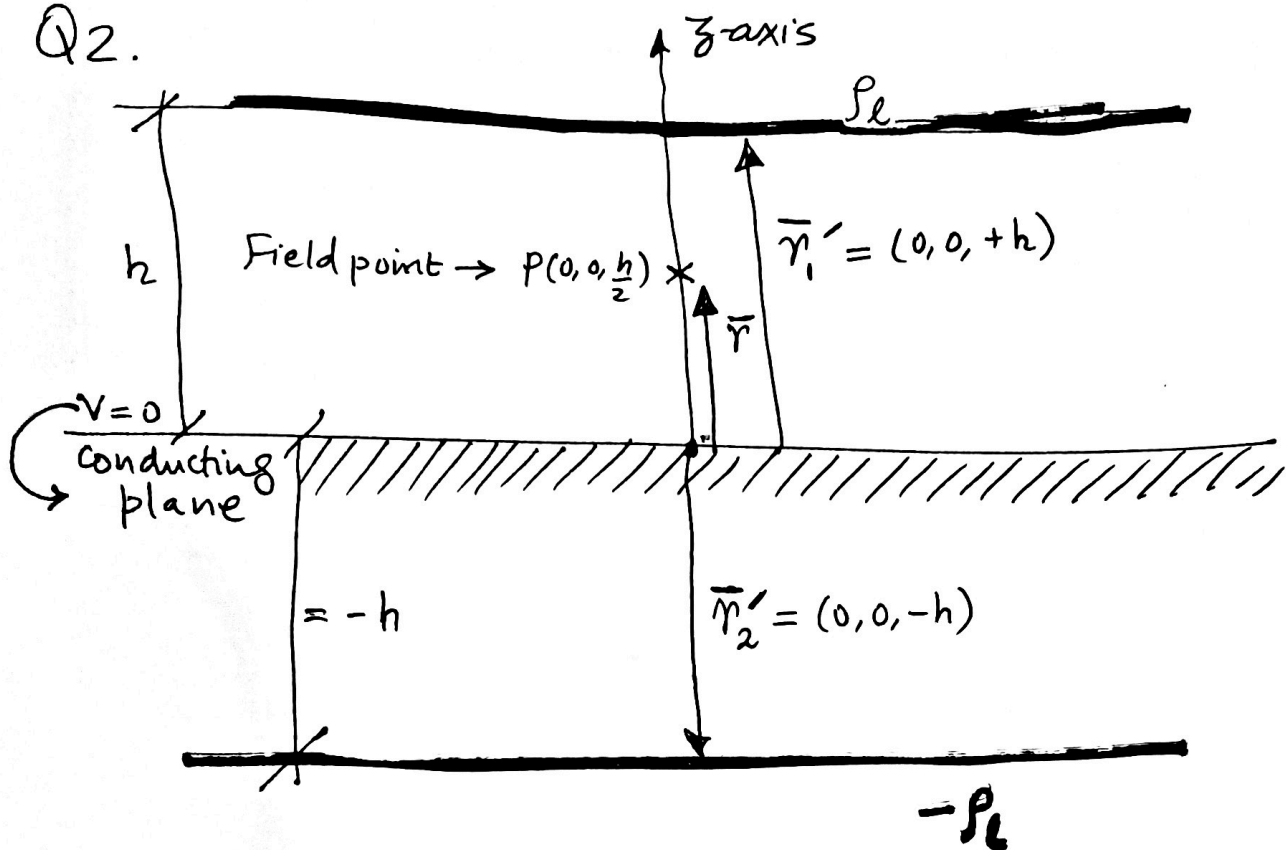


Midterm II (Retake Solution)

①

Q2.



Field Intensity due to line charge and its image.

$$\begin{aligned}\vec{E}_{\text{line(net)}} &= \vec{E}_{\text{line(1)}} + \vec{E}_{\text{line image}} \\ &= \frac{\rho_L}{2\pi\epsilon_0\rho_1} \hat{a}_{\rho_1} + \frac{-\rho_L}{2\pi\rho_2\epsilon_0} \hat{a}_{\rho_2}\end{aligned}$$

$$\vec{\rho}_1 = (0, 0, \frac{h}{2}) - (0, 0, h) = \vec{r} - \vec{r}'_1 = -\frac{h}{2} \hat{a}_z$$

$$\text{So } \hat{a}_{\rho_1} = -\hat{a}_z$$

$$\rho_1 = |\vec{\rho}_1| = \frac{h}{2}$$

$$\vec{\rho}_2 = (0, 0, h/2) - (0, 0, -h) = \vec{r} - \vec{r}'_2 = \frac{3h}{2} \hat{a}_z$$

$$\text{So } \hat{a}_{\rho_2} = +\hat{a}_z$$

$$\rho_2 = |\vec{\rho}_2| = \frac{3h}{2}$$

$$\vec{E}_{\text{line (net)}} = \frac{(\rho_L)(-a_z)}{2\pi\epsilon_0(h/2)} + \frac{(-\rho_L)(a_z)}{2\pi\epsilon_0(3h/2)} \quad (2)$$

$$= -\frac{\rho_L a_z}{\pi\epsilon_0 h} - \frac{\rho_L a_z}{3\pi\epsilon_0 h}$$

$$= -\frac{\rho_L}{\pi\epsilon_0 h} \left(1 + \frac{1}{3}\right) a_z$$

$$= -\frac{4\rho_L}{3\pi\epsilon_0 h} a_z$$

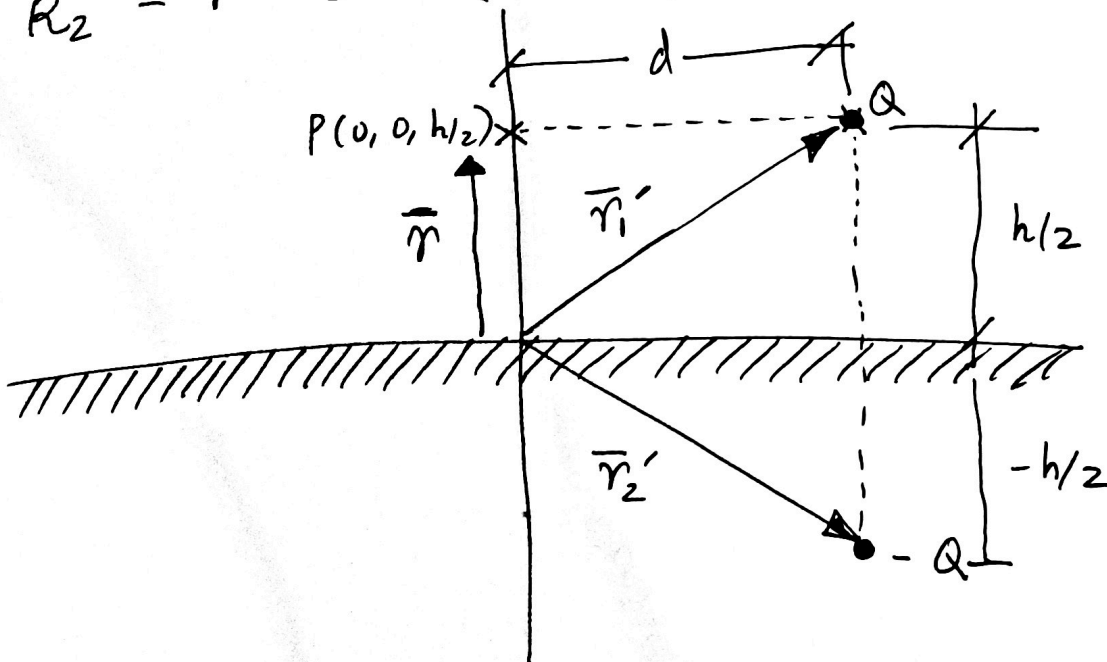
Field intensity due to point charge and its image

$$\vec{E}_{\text{point (net)}} = \vec{E}_{\text{point}} + \vec{E}_{\text{point image}}$$

$$= \frac{Q \vec{R}_1}{4\pi\epsilon_0 R_1^3} + \frac{(-Q) \vec{R}_2}{4\pi\epsilon_0 R_2^3}$$

$$\vec{R}_1 = \vec{r} - \vec{r}_1' = (0, 0, h/2) - (0, d, h/2) = -d a_y$$

$$\vec{R}_2 = \vec{r} - \vec{r}_2' = (0, 0, h/2) - (0, d, -h/2) = -d a_y + h a_z$$



$$\begin{aligned}\vec{E}_{\text{point (net)}} &= \frac{Q(-d\hat{y})}{4\pi\epsilon_0(d^3)} + \frac{(-Q)(-d\hat{y} + h\hat{z})}{4\pi\epsilon_0(d^2 + h^2)^{3/2}} \quad (3) \\ &= \frac{(-Q)\hat{y}}{4\pi\epsilon_0 d^2} + \frac{(-Q)(-d\hat{y} + h\hat{z})}{4\pi\epsilon_0(d^2 + h^2)^{3/2}}\end{aligned}$$

Total Field:

$$\vec{E}_{\text{net}} = \vec{E}_{\text{point(net)}} + \vec{E}_{\text{line(net)}}$$

Similarly, we find Electric potential.

$$V = V_+^{\text{line}} + V_-^{\text{line}} + V_+^{\text{point}} + V_-^{\text{point}}$$

$$= -\frac{\rho_L \ln\left(\frac{h}{2}\right)}{2\pi\epsilon_0} - \frac{(-\rho_L) \ln\left(\frac{3h}{2}\right)}{2\pi\epsilon_0}$$

$$+ \frac{Q}{4\pi\epsilon_0 d} + \frac{-Q}{4\pi\epsilon_0(d^2 + h^2)^{1/2}}$$

$$= \frac{\rho_L}{2\pi\epsilon_0} \left[\ln\left(\frac{3h}{2}\right) - \ln\left(\frac{h}{2}\right) \right] + \frac{Q}{4\pi\epsilon_0 d} - \frac{Q}{4\pi\epsilon_0(d^2 + h^2)^{1/2}}$$

$$V = \frac{\rho_L}{2\pi\epsilon_0} \ln(3) + \frac{Q}{4\pi\epsilon_0 d} \left(1 - \frac{d}{\sqrt{d^2 + h^2}} \right).$$

This brief solution relies on the fact that

$$V = - \int \vec{E} \cdot d\vec{l}.$$

Q1. $\nabla^2 V = 0 \Rightarrow V = A \ln \rho + B$

This is a brief solution, where not all steps are shown.

(u) There are two regions, Therefore

$$V_1 = A_1 \ln \rho + B_1$$

$$V_2 = A_2 \ln \rho + B_2$$

The boundary conditions are

$$V_1(\rho=a) = 0 \Rightarrow A_1 \ln a + B_1 = 0$$

$$V_2(\rho=c) = 0 \Rightarrow A_2 \ln c + B_2 = 0$$

$$V_1(\rho=b) = V_2(\rho=b) \Rightarrow A_1 \ln b + B_1 = A_2 \ln b + B_2$$

we also have at $\rho=b$.

$$D_{1n} - D_{2n} = \rho_s$$

$$\epsilon_1 E_1 - \epsilon_2 E_2 = \rho_s \quad (\text{at } \rho=b)$$

$$E_1 = -\frac{\partial V_1}{\partial \rho} = -\frac{A_1}{\rho}$$

$$E_2 = -\frac{\partial V_2}{\partial \rho} = -\frac{A_2}{\rho}$$

$$\epsilon_1 \frac{A_1}{b} - \epsilon_2 \frac{A_2}{b} = \rho_s \Rightarrow \epsilon_1 A_1 - \epsilon_2 A_2 = b \rho_s$$

$$B_1 = -A_1 \ln a$$

$$B_2 = -A_2 \ln c$$

$$A_1 \ln b - A_1 \ln a = A_2 \ln b - A_2 \ln c$$

$$A_1 \ln\left(\frac{b}{a}\right) = A_2 \ln\left(\frac{b}{c}\right)$$

$$\epsilon_1 A_1 - \epsilon_2 A_2 = b f_s$$

$$\ln\left(\frac{b}{a}\right) A_1 - \ln\left(\frac{b}{c}\right) A_2 = 0$$

(5)

$$\ln\left(\frac{b}{a}\right) \left(\frac{b f_s + \epsilon_2 A_2}{\epsilon_1} \right) - \ln\left(\frac{b}{c}\right) A_2 = 0$$

$$\ln\left(\frac{b}{a}\right) b f_s + \epsilon_2 \ln\left(\frac{b}{a}\right) A_2 - \epsilon_1 \ln\left(\frac{b}{c}\right) A_2 = 0$$

$$A_2 = \frac{-\ln\left(\frac{b}{a}\right) b f_s}{\epsilon_2 \ln\left(\frac{b}{a}\right) - \epsilon_1 \ln\left(\frac{b}{c}\right)}$$

$$A_2 = \frac{b f_s}{\frac{\ln(b/c) \epsilon_1 - \epsilon_2}{\ln(b/a)}}$$

$$\ln\left(\frac{b}{a}\right) A_1 - \ln\left(\frac{b}{c}\right) \left(\frac{\epsilon_1 A_1 - b f_s}{\epsilon_2} \right) = 0$$

$$\epsilon_2 \ln\left(\frac{b}{a}\right) A_1 - \epsilon_1 \ln\left(\frac{b}{c}\right) A_1 + \ln\left(\frac{b}{c}\right) b f_s = 0$$

$$A_1 = \frac{\ln\left(\frac{b}{c}\right) b f_s}{\ln\left(\frac{b}{c}\right) \epsilon_1 - \ln\left(\frac{b}{a}\right) \epsilon_2}$$

Once A_1 & A_2 are computed $V_1 = A_1 \ln r + B_1$,

where $B_1 = -A_1 \ln a$.

& $V_2 = A_2 \ln r + B_2$,

where $B_2 = -A_2 \ln c$.

$$\bar{E}_1 = -\frac{A_1}{r} 2r$$

$$\& \bar{E}_2 = -\frac{A_2}{r} 2r$$